CBSE Test Paper 03

Chapter 5 Laws of Motion

1. Inertia refers to **1**
    a. resistance to change
    b. dullness
    c. ease of motion
    d. slow motion
2. A truck starts from rest and accelerates uniformly at 2.0 m s^{-2}. At t = 10 s, a stone is dropped by a person standing on the top of the truck (6 m high from the ground). What is the magnitude of acceleration (in m s^{-2}) of the stone at t = 11s? (Neglect air resistance.) **1**
    a. 14
    b. 12
    c. 8
    d. 10
3. Impulse is a **1**
    a. vector and equals the change of mass
    b. vector and equals the change of momentum
    c. vector and equals the inertia of momentum
    d. scalar and equals the change of mass
4. A batsman deflects a ball by an angle of 45° without changing its initial speed which is equal to 54 km/h. What is the impulse imparted to the ball? (Mass of the ball is 0.15 kg.) **1**
    a. 4.4 kg m s^{-1}
    b. 4.8 kg m s^{-1}
    c. 4.6 kg m s^{-1}
    d. 4.2 kg m s^{-1}

5. A man of mass 70 kg stands on a weighing scale in a lift which is freely falling under gravity. What would be the reading on the scale? **1**

   a. 0 kg
   b. 125 kg
   c. 95 kg Hz
   d. 115 kg

6. Two blocks of masses m_1 and m_2 are connected at the two ends of a light spring kept on a smooth horizontal surface. The two masses are pulled apart by two forces F_1 and F_2 respectively and then released. Prove that the ratio of their acceleration is inversely proportional to their masses. **1**

7. A person of mass 50 kg stands on a weighing scale on a lift. If the lift is descending with a downward acceleration of 9 ms^{-2} what would be the reading of the weighing scale? (g = 10 ms^{-2}) **1**

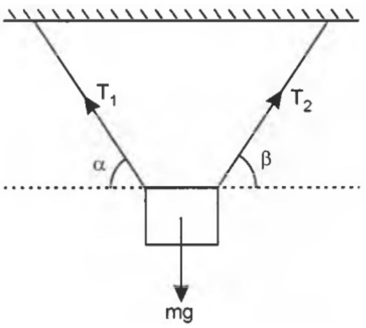
8. A thief jumps from the roof of a house with a box of weight W on his head. What will be the weight of the box as experienced by the thief during jump? **1**

9. Force of 16N and 12N are acting on a mass of 200 kg in mutually perpendicular directions. Find the magnitude of the acceleration produced? **2**

10. A shell of mass 0.020 kg is fired by a gun of mass 100 kg. If the muzzle speed of the shell is 80 ms^{-1}, what is the recoil speed of the gun? **2**

11. State Newton's first law of motion. Give an example to illustrate it. **2**

12. A body of mass m is suspended by two strings making angles α and β with the horizontal as shown in Figure. Calculate the tensions in the two strings. **3**



13. There are three forces $\vec{F}_1, \vec{F}_2$ and $\vec{F}_3$ acting on a body, all acting on a point P on the body. The body is found to move with uniform speed. **3**

   a. Show that the forces are coplanar.
   b. Show that the torque acting on the body about any point due to these three forces is zero.

14. A monkey of mass 40 kg climbs on a rope which can stand a maximum tension 600 N. In which of the following cases will the rope break? **3**
   The monkey

   a. climbs up with an acceleration of $6 m/s^2$
   b. climbs down with an acceleration of $4 m/s^2$
   c. climbs up with a uniform speed of 5m/s
   d. falls down the rope freely under gravity. Take $g = 10 m/s^2$ and ignore the mass of the rope.

15. Consider a ball falling from a height of 2 m and rebounding to a height of 0.5 m. If the mass of the ball is 60 g, find the impulse and the average force between the ball and the ground. The time for which the ball and the ground remained in contact was 0.2 s. **5**